

Overview



A1 M12 Light Joystick	A11 Brass Pillow M2.5x4 - 2pcs	B1 Main Board
A2 1RBLK Button Knob - 2 pcs	A12 Nut M2.5 - 2pcs	B2 Connectors Board
A3 Screws PH1 M2.5x6 - 4 pcs	A13 OEJNI-90-4-7 Knob	B3 Encoders Board
A4 Screws PH1 M2.5x16 - 6 pcs	A14 Pillow M6x22	P1 3D Body Top
A5 Screws PH1 M2x4 - 4 pcs	A15 Bearing M6x12x4	P2 3D Body Bottom
A6 OEJNI-50-4-7 Knob - 2 pcs	A16 Washer M6x12x0.5 - 2pcs	P3 3D Body Display Stand
A7 OEJL-50-4-5 Knob	A17 Oil Seal M6x12x3	P4 3D Body Pillow Joint
A8 Nut M6 - 3 pcs	A18 Magnet M6x3	
A9 LCD TFT 135x240 1.14 inch	A19 FFC 10-pin B-dir	
A10 Screw PH1 M2.5x4 - 2pcs.	A20 FFC 8-pin A-dir	

Figure 1. The MiniRemote kit contents

Notes:

- The encoder magnet must not be heated. Exposure to high temperatures causes it to lose its magnetic properties;
- When installing FFC cables into connectors, you can use an auxiliary tool, such as a plastic hand tool. Do not use sharp metal objects for this purpose;
- If the joystick is installed upside down, you can invert the axis back using software mixers (Remote → Mixer Settings);
- When threading plastic with a screw, it may feel quite tight the first time. During subsequent assembly/disassembly, the screw rotation will become much smoother;
- The nuts **A8** are included in the set and are attached to their corresponding radio components. Remove them before step 8.

Assembly order

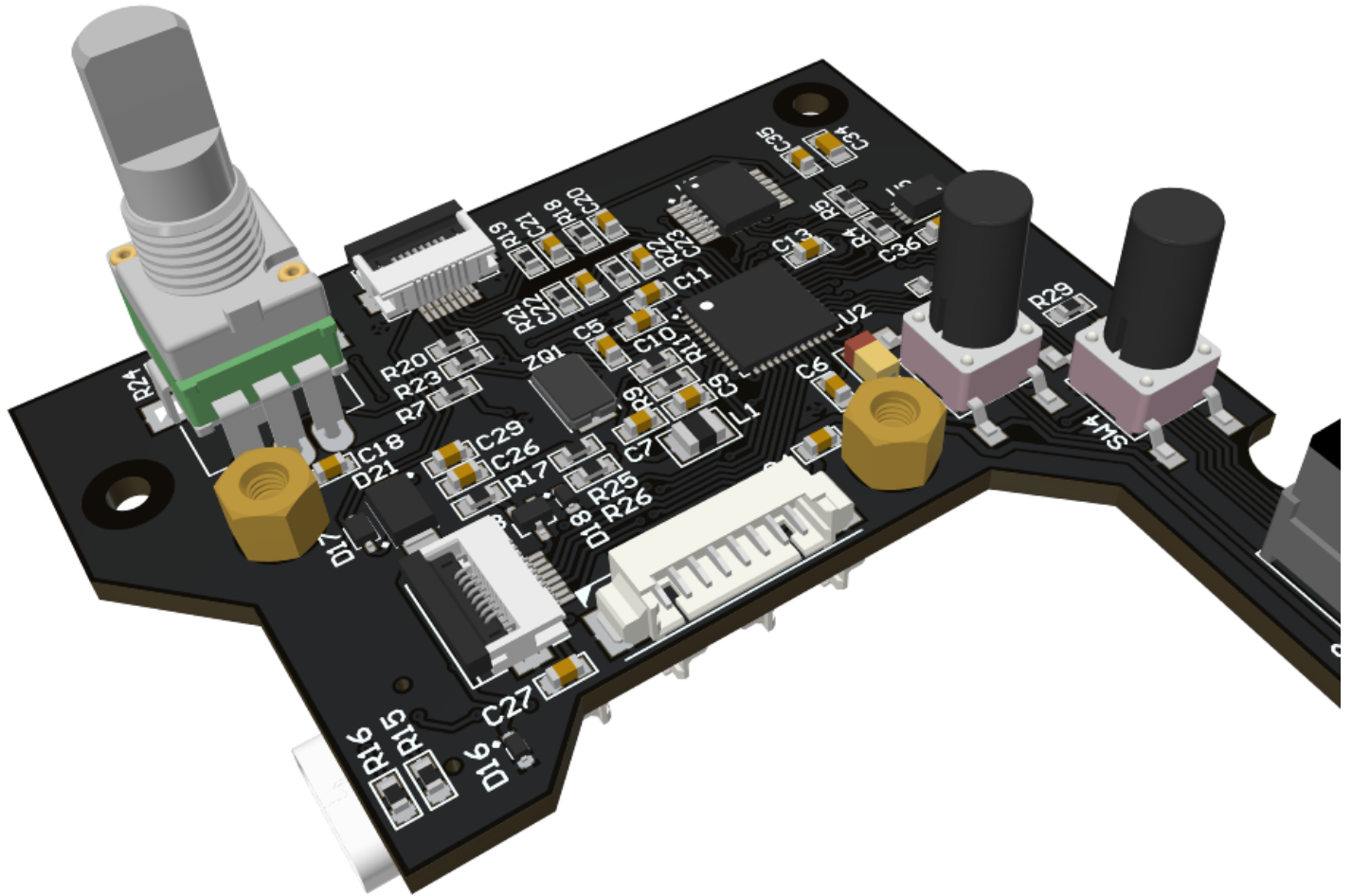


Figure 2. Installation of standoffs and button actuators

1. To prepare the motherboard, first install two standoffs for the display using M2.5 nuts **A12** and M2.5x4 brass pillows **A11**. Installing washers is optional and serves only to preserve the solder mask layer on the board. Avoid overtightening the nuts to prevent damage to the motherboard. Then, attach the 1RBLK actuators **A2** to the SW1 and SW4 buttons.

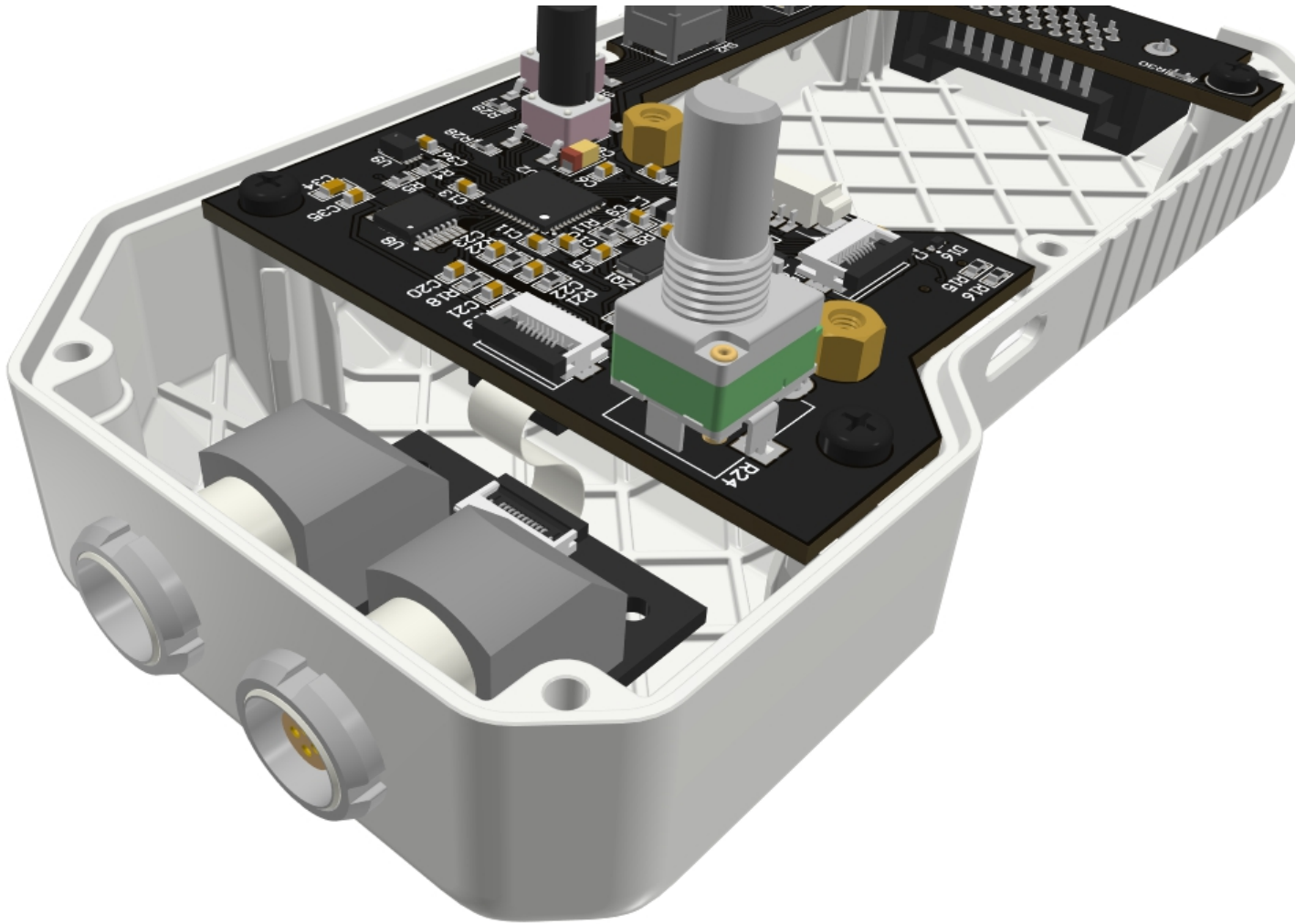


Figure 3. Installation of the board with connectors

2. First, remove the front locking nuts from the connectors. The connection is made using a 10-pin 0.5mm FFC cable, 30mm in length **A19**. The corresponding connector on the main board is located on its underside. Secure both ends of the cable tightly into the connectors on these boards. Next, take the bottom part of the remote control case **P2** and install the assembly of interconnected boards into it. Loosen the rear nuts on the LEMO connectors to about halfway along their thread length. It is recommended to start by installing this section into the front part of the remote control case. Then, place the main board onto the supports located on the side walls of the case. Ensure that all four mounting holes and the side USB Type-C connector are properly aligned. Finally, secure the board with the LEMO connectors using the previously removed external nuts, and attach the main board with four M2.5x6 screws **A3**.

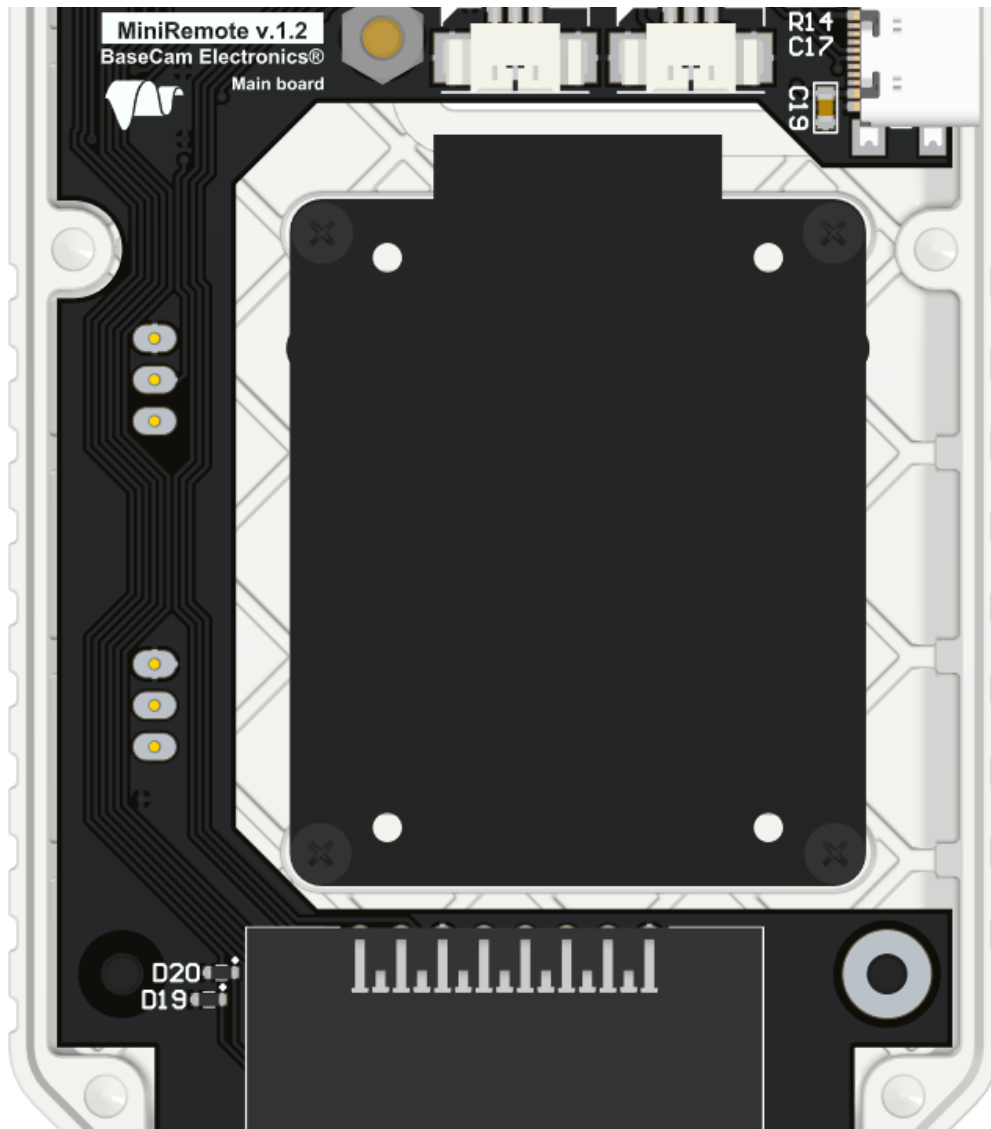


Figure 4. Joystick installation

3. Take the top part of the MiniRemote case **P1** and secure the joystick **A1** using four M2x4 self-tapping screws **A5**. It is recommended to install the joystick so that the Hall sensor of the X-axis (ROLL) faces the back of the MiniRemote. Avoid overtightening the screws. Then insert the joystick plug into the corresponding connector on the top of the motherboard, next to the joystick.

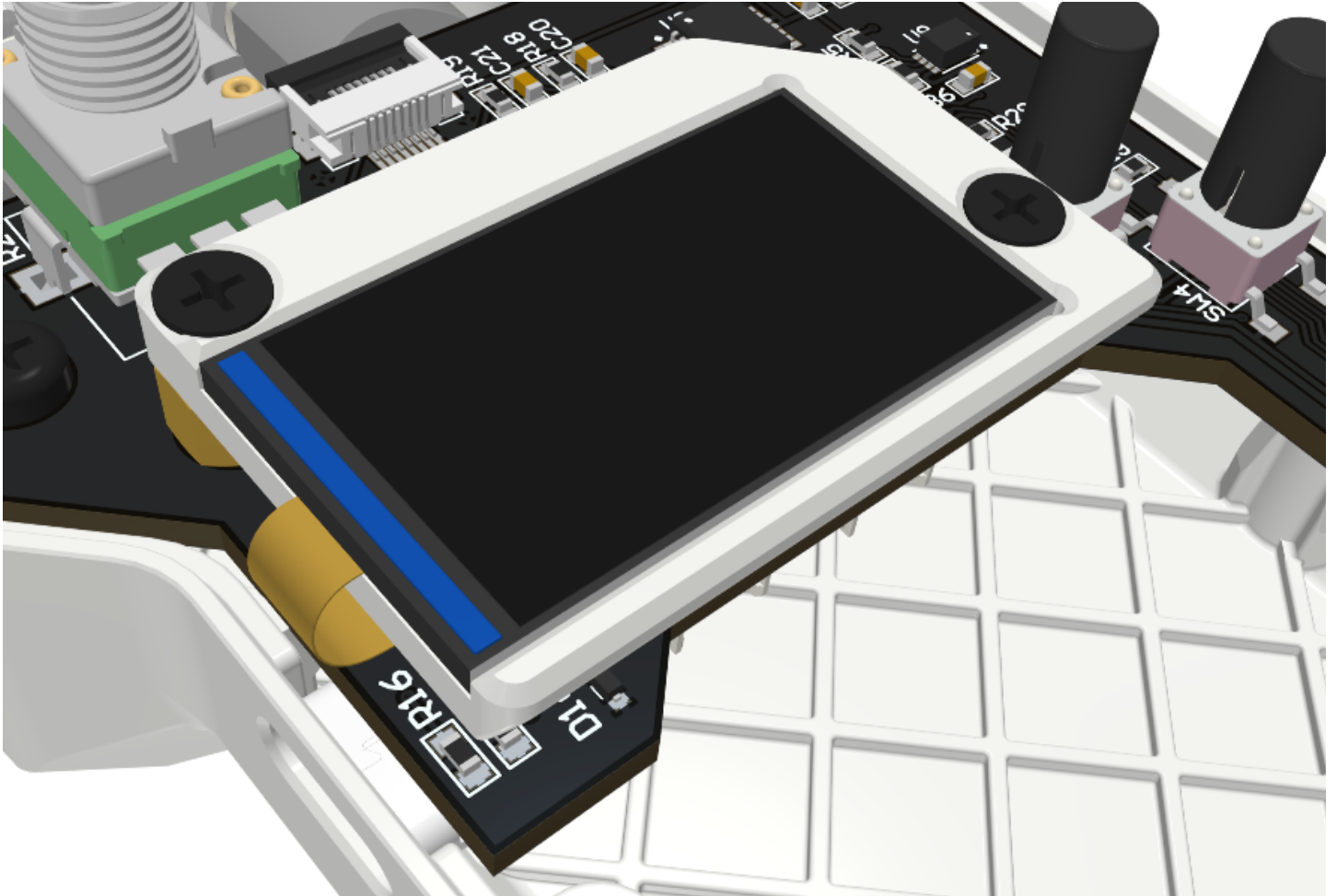


Figure 5. Display installation

4. First, insert the display **A9** into the FPC connector as shown in the diagram above. Then position the display frame **P3** on the previously secured brass pillows and secure with screws **A10**. Place the display itself inside this frame. Optionally, you can use thin double-sided tape or super glue for additional fixation. Carefully bend the display flex cable inward so that it does not extend beyond the boundaries of the remote control case.

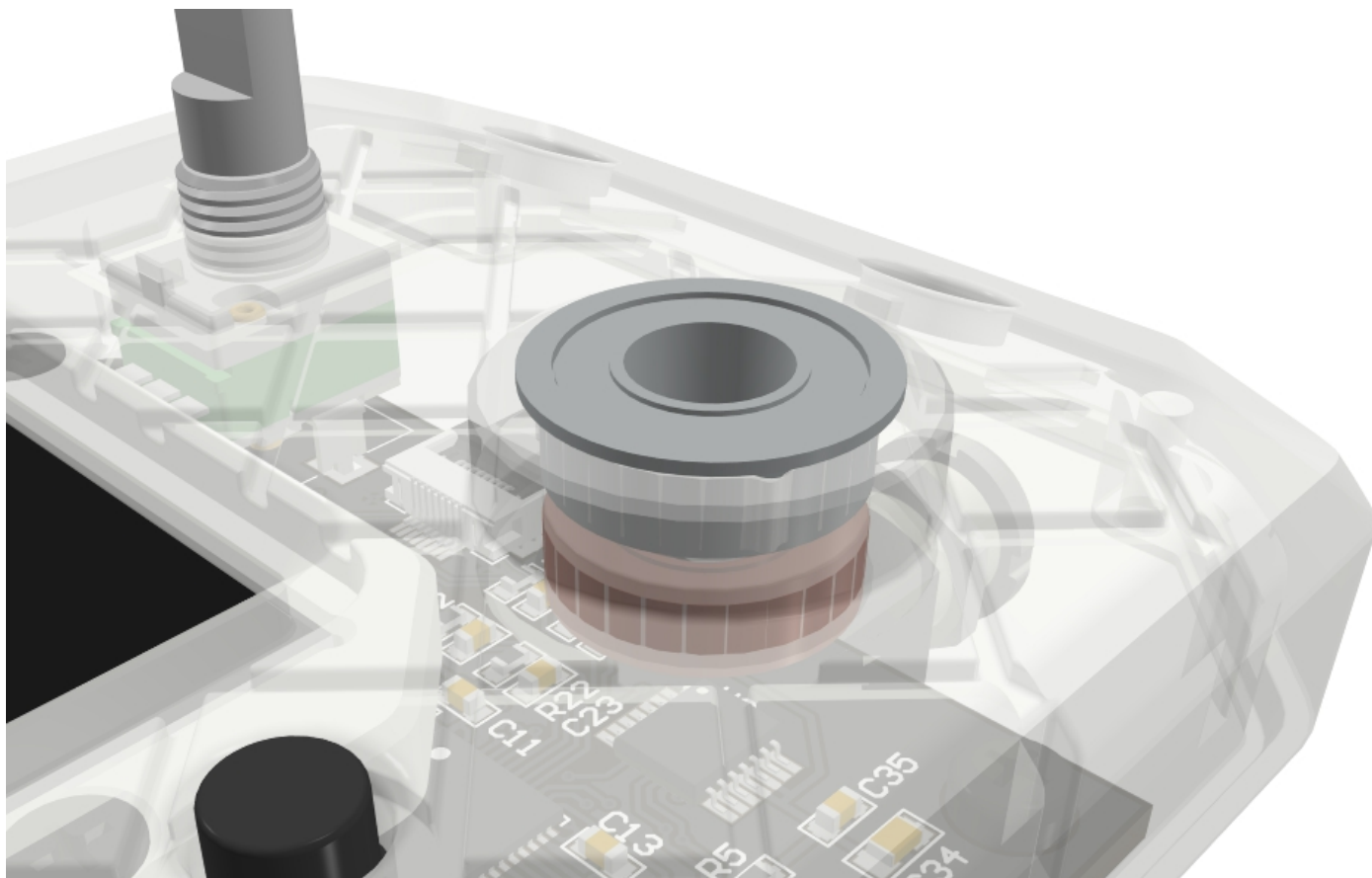


Figure 6. Encoder knob installation, part 1

5. Secure the encoder's seal **A17** and bearing **A15** in the designated holes, as shown in the diagram above. They will hold firmly in place, but optionally, you can add a small amount of super glue during installation for extra security.

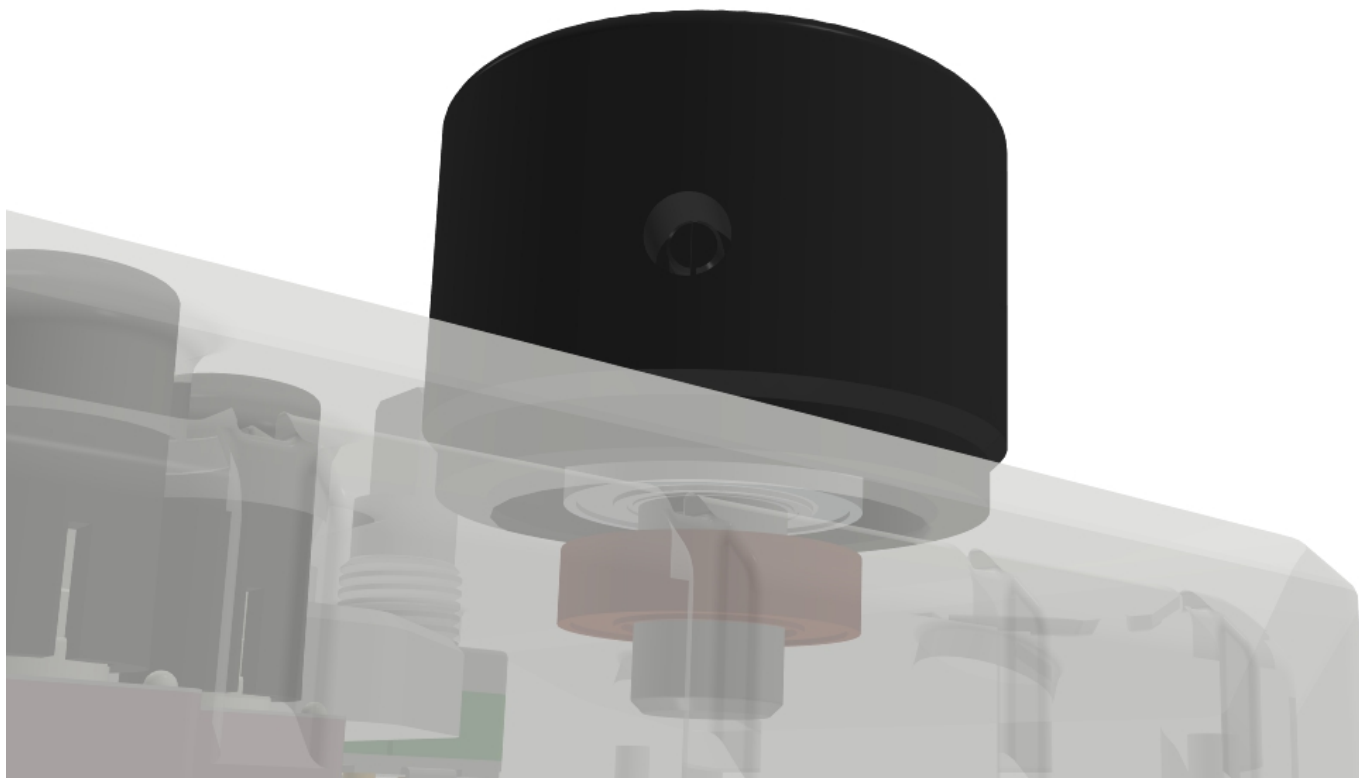


Figure 7. Encoder knob installation, part 2

6. Now, prepare the encoder knob **A13** for installation. Attach the knob to the shaft **A14** using the side screw. Then, from the bottom side, place a thin nylon washer M6x12x0.5 **A16** onto the shaft. Lubricate the shaft evenly around its circumference. It can now be inserted through the bearing and seal into the case.

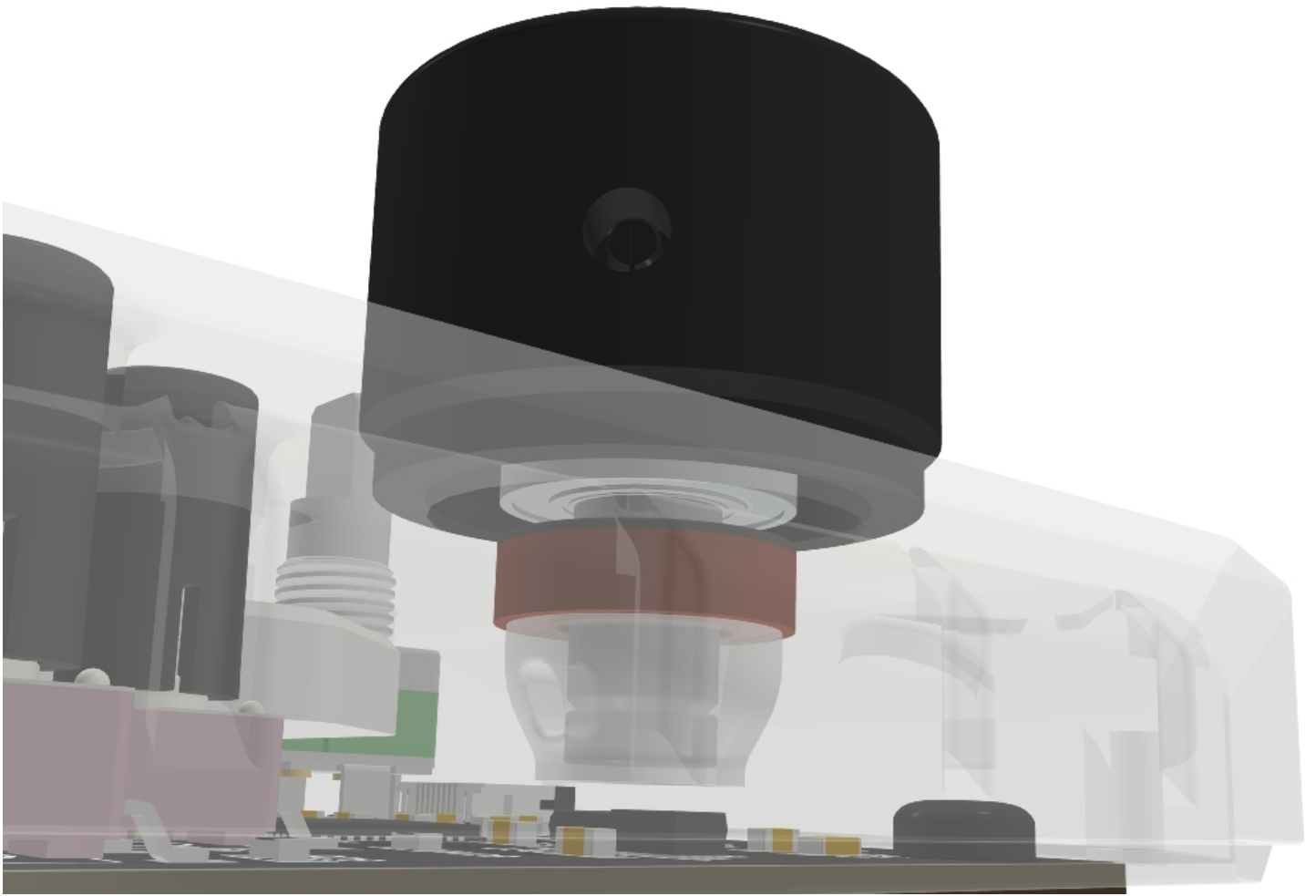


Figure 8. Encoder knob installation, part 3

7. Slide a nylon washer M6x12x0.5 **A16** onto the shaft. Insert an M6x3 magnet **A18** into the Pillow Joint component **P4**. Then press this component onto the shaft from above, as shown in the picture above.

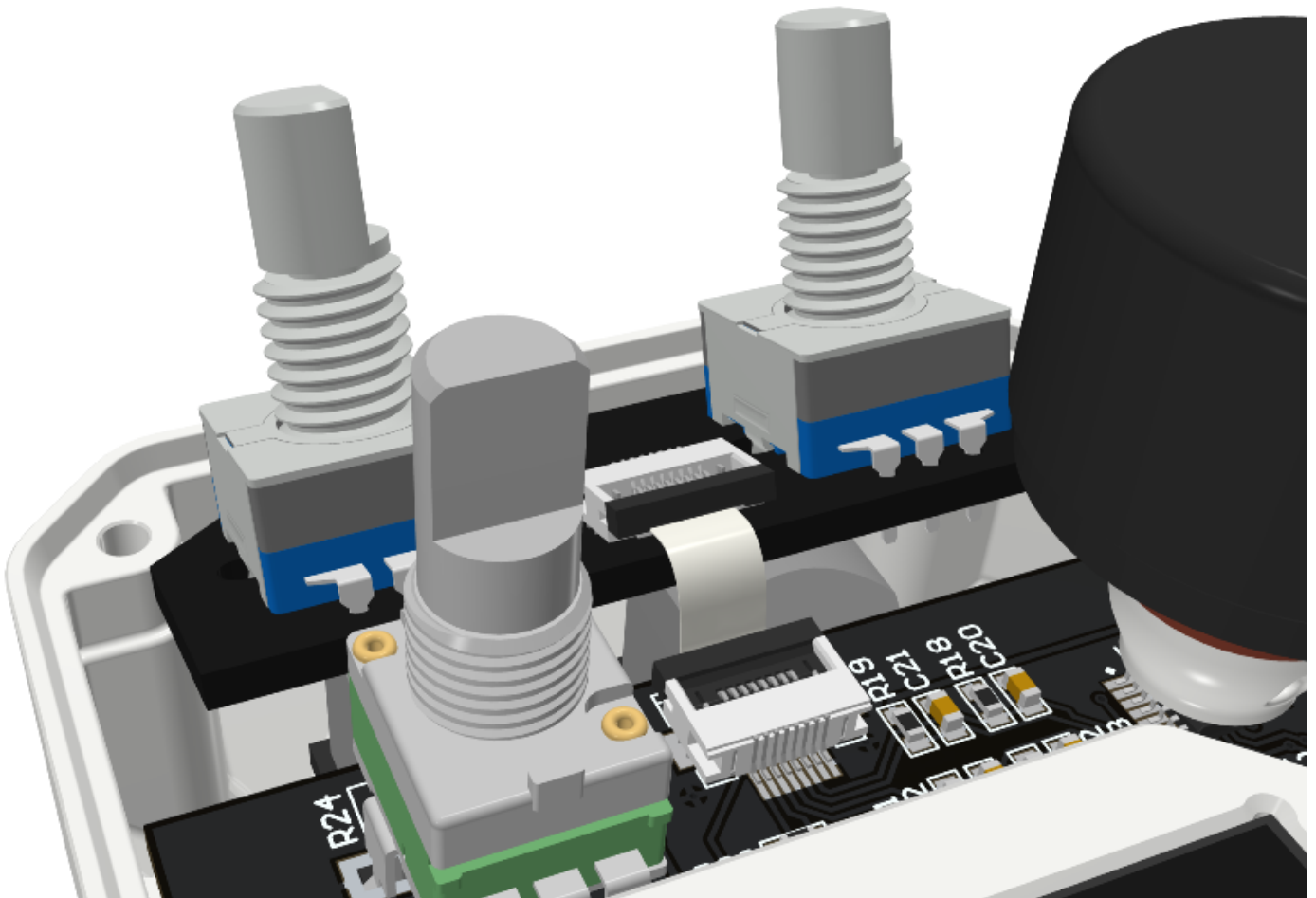


Figure 9. Encoders board installation

8. The final internal component is the encoder board **B3**. It needs to be connected to the motherboard using an FFC cable **A20**. Now you can join the top and bottom halves of the remote control. While doing this, try to support the encoder board from underneath first, and then from above.



Figure 10. Knobs installation

9. Now secure the encoders and the variable resistor to the case using M6 nuts **A8**. Then attach the knobs OEJL-50-4-5 **A7** and OEJNI-50-4-7 **A6** (the one with the white mark goes on the resistor). Fix them in place using the side screw. Ensure they are positioned at a height from the case surface such that the encoder buttons can be pressed freely and the variable resistor knob does not rub against the plastic when rotated.



Figure 11. Securing the top and bottom parts of the case

10. Flip the remote control over and secure the two halves of the case with six screws **A4**. The device is now fully assembled and ready for use.